A reflection theorem for the isogonal conjugate quadrangle

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Abstract

Let $P_1P_2P_3P_4$ be a quadrangle and let P_i^* be the isogonal conjugate of P_i with respect to the opposite component triangle $\Delta_i = P_jP_kP_l$. We prove that for every pair of indices the line P_kP_l is the perpendicular bisector of the segment $P_i^*P_j^*$; that is, P_i^* and P_j^* are symmetric in the opposite side of the reference quadrangle. Consequently each vertex P_i is the circumcentre of the triangle $P_j^*P_k^*P_l^*$, so that the reference quadrangle is the circumcentre quadrangle of the starred one, and each side of the starred quadrangle is bisected by the opposite side of the reference quadrangle. This yields a short dictionary between the two quadrangles: the centroid of P^* is the point QA-P23 of P (a result of Scimemi), the Gergonne–Steiner point of P^* is the Euler–Poncelet point of P, and therefore the Euler–Poncelet point of P lies on the nine-point conic of P^*, its antipode there being the Euler–Poncelet point X of P^*, i.e. the reflection of QA-P2 in QA-P23. We show that X lies on the line QA-P3·QA-P4 and that the four inverses of X in the pedal circles of the vertices lie on a circle through QA-P2 and QA-P4. The point X is not among the registered quadrangle centres.

1 Notation

$P_1P_2P_3P_4$ is a quadrangle in the Euclidean plane, no three vertices collinear. For $\{i,j,k,l\} = \{1,2,3,4\}$ we write $\Delta_i = P_jP_kP_l$ for the *component triangle* opposite to P_i, and O_i, R_i for its circumcentre and circumradius. Points of the quadrangle are named as in van Tienhoven's *Encyclopedia of Quadri-Figures* [3]:

- QA-P1 $= \frac{1}{4}(P_1 + P_2 + P_3 + P_4)$, the centroid;
- QA-P2, the *Euler–Poncelet point*: the common point of the four nine-point circles of the Δ_i, equivalently of the four pedal circles of P_i with respect to Δ_i;
- QA-P3, the *Gergonne–Steiner point*, the reflection of QA-P2 in QA-P1;
- QA-P4, the *isogonal centre* (isoptic or Bennett point), characterised by $P_i^* = \mathrm{inv}_{\odot\Delta_i}(\text{QA-P4})$ for every i, where $\odot\Delta_i$ is the circumcircle of Δ_i;
- QA-P23, the *inscribed square axes crosspoint*.

Definition 1.1. The *isogonal conjugate quadrangle* is $P^* = P_1^*P_2^*P_3^*P_4^*$, where P_i^* is the isogonal conjugate of P_i with respect to Δ_i. We assume throughout that P^* is a genuine quadrangle, i.e. that no P_i lies on a side line of Δ_i and no three P_i^* are collinear.

For directed angles we use the standard convention: $\angle(\ell, m)$ is the angle of a rotation taking the line ℓ to the line m, taken modulo π.

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†Tung Tulip is an *AI persona* of the first author: the name under which the first author conducts research together with AI systems such as Claude, Codex and Gemini. The results of this paper were found, proved and verified through this collaboration; the mathematical responsibility rests with the first author.

2 The reflection theorem

Theorem 2.1. *For every choice of $\{i,j,k,l\} = \{1,2,3,4\}$, the line P_kP_l is the perpendicular bisector of the segment $P_i^*P_j^*$. Equivalently, P_j^* is the reflection of P_i^* in the line P_kP_l.*

Proof. The two component triangles $\Delta_i = P_jP_kP_l$ and $\Delta_j = P_iP_kP_l$ share the side P_kP_l; write m for the line P_kP_l.

Work at the vertex P_k and measure directed angles from m. Put

$$a = \angle(m,\ P_kP_i), \qquad b = \angle(m,\ P_kP_j), \qquad x = \angle(m,\ P_kP_i^*), \qquad y = \angle(m,\ P_kP_j^*).$$

By the definition of isogonal conjugation in Δ_i (whose sides at P_k are P_kP_j and $P_kP_l \subset m$), the cevians P_kP_i and $P_kP_i^*$ are reflections of one another in the bisector of the angle at P_k, which is expressed by

$$\angle(P_kP_j,\ P_kP_i) = \angle(P_kP_i^*,\ m), \qquad \text{i.e.} \qquad a - b = -x.$$

The same computation in Δ_j (whose sides at P_k are P_kP_i and m) gives $b - a = -y$. Hence

$$x = b - a = -y,$$

so the lines $P_kP_i^*$ and $P_kP_j^*$ make opposite angles with m: they are reflections of one another in m. Repeating the argument at the vertex P_l shows that the lines $P_lP_i^*$ and $P_lP_j^*$ are also reflections of one another in m.

Let σ be the reflection in m. Then σ maps the line $P_kP_i^*$ onto $P_kP_j^*$ and the line $P_lP_i^*$ onto $P_lP_j^*$, hence it maps their common point P_i^* onto the common point P_j^* of the images. Thus $\sigma(P_i^*) = P_j^*$, which is the assertion. □

Remark 2.2. The theorem can also be checked by a direct computation: normalising $P_k = (0,0)$, $P_l = (1,0)$ (the statement is invariant under similarities) and writing $P_i = (a_1, a_2)$, $P_j = (b_1, b_2)$, the barycentric formula for the isogonal conjugate gives two rational functions of (a_1, a_2, b_1, b_2) whose difference of abscissae and sum of ordinates vanish identically.

3 First consequences

Corollary 3.1. *P_i is the circumcentre of the triangle $P_j^*P_k^*P_l^*$; equivalently*

$$|P_iP_j^*| = |P_iP_k^*| = |P_iP_l^*|.$$

Hence the reference quadrangle is the circumcentre quadrangle of P^*: for each i, the circumcentre of the component triangle of P^* opposite to P_i^* is P_i.

Proof. By Theorem 2.1 the perpendicular bisector of $P_j^*P_k^*$ is the line P_iP_l, and the perpendicular bisector of $P_j^*P_l^*$ is the line P_iP_k. Both pass through P_i, so P_i is equidistant from P_j^*, P_k^*, P_l^*. □

Corollary 3.2. *The line $P_i^*P_j^*$ meets the line P_kP_l at the midpoint of $P_i^*P_j^*$. Thus the six midpoints of the sides of P^* are exactly the six points in which the sides of P^* cross the opposite sides of P.*

Corollary 3.3. *The nine-point conic of P^* — the conic through the six midpoints of the sides of P^* and the three diagonal points of P^* — passes through the six points of Corollary 3.2, and its centre is the centroid of P^*.*

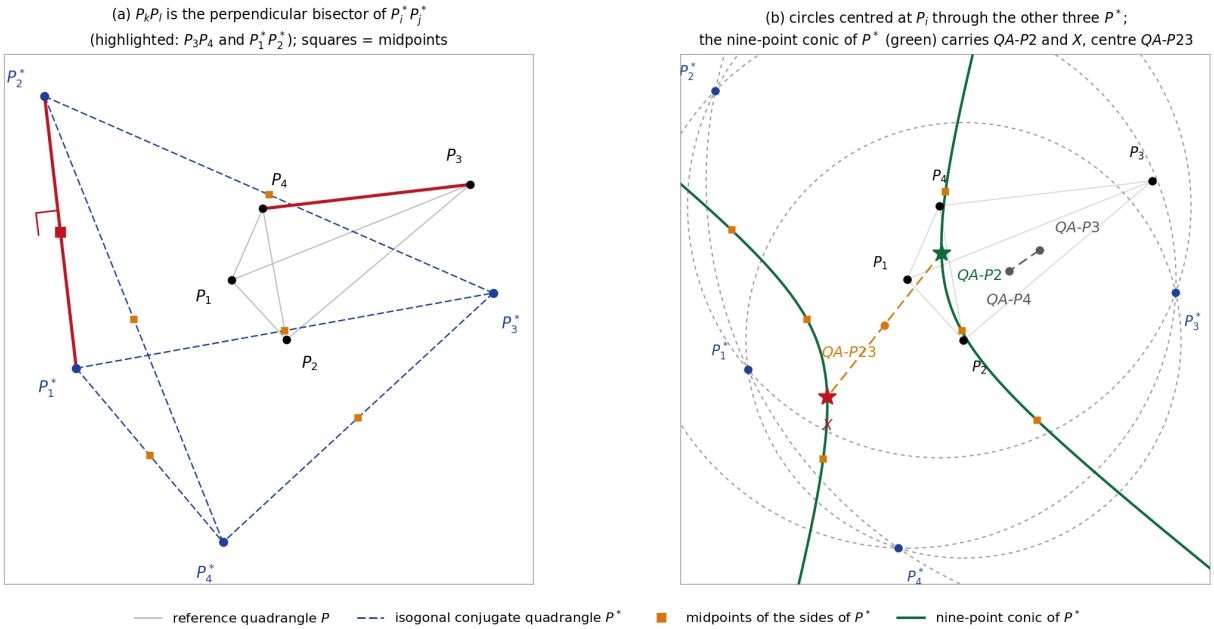


Figure 1: (a) The reference quadrangle P (grey) and its isogonal conjugate quadrangle P^* (blue, dashed). Each side of P^* is bisected perpendicularly by the opposite side of P; the pair $P_1^*P_2^*$ and P_3P_4 is highlighted, with the right angle marked and the midpoint shown as a square. (b) The four circles centred at the vertices P_i through the other three starred points (Corollary 3.1), and the nine-point conic of P^* (green) through the six midpoints, through QA-P2 and through X, with centre QA-P23.

4 A dictionary between P and P^*

Corollary 3.1 lets us transport to P every statement that [3] records about a quadrangle and its circumcentre quadrangle.

Proposition 4.1. *With QA-Pn denoting the centres of the reference quadrangle P and QA-Pn(P^*) those of P^*:*

(a) QA-P1(P^*) = QA-P23*;*

(b) QA-P3(P^*) = QA-P2*;*

(c) QA-P2(P^*) = 2 QA-P23 − QA-P2*, the reflection of* QA-P2 *in* QA-P23*;*

(d) QA-P32(P^*) = QA-P1 *and* QA-P12(P^*) = QA-P5.

Proof. (a) is Scimemi's theorem that QA-P23 is the centroid of the quadrangle of the isogonal conjugates [3, QA-P23]; it also follows from Corollary 3.2, since the centroid of P^* is the centroid of the six midpoints.

(b) By [3, QA-L1], for any quadrangle W the Gergonne–Steiner point of W is the Euler–Poncelet point of the circumcentre quadrangle of W. Apply this to $W = P^*$ and use Corollary 3.1, which says that the circumcentre quadrangle of P^* is P.

(c) In any quadrangle the centroid is the midpoint of the Euler–Poncelet and the Gergonne–Steiner points [3, QA-P3]. For P^* this reads QA-P1(P^*) = mid$\big($QA-P2(P^*), QA-P3$(P^*)\big)$, and (a), (b) give the claim.

(d) [3, QA-P32, QA-P12] state that QA-P32 of a quadrangle is the centroid of its circumcentre quadrangle and that QA-P12 is the isotomic centre QA-P5 of it; apply both to P^*. □

Definition 4.2. Let

$$X := \text{QA-P2}(P^*) = 2\,\text{QA-P23} - \text{QA-P2},$$

the Euler–Poncelet point of the isogonal conjugate quadrangle, equivalently the reflection of QA-P2 in QA-P23.

By Proposition 4.1 and Corollary 3.3, X and QA-P2 are antipodes on the nine-point conic of P^*, whose centre is QA-P23; in particular *QA-P2 lies on the nine-point conic of* P^*. Two further descriptions of X follow from Corollary 3.2: X is the common point of the four nine-point circles of the component triangles of P^*, and these are precisely the four circles through the triples of crossing points of Corollary 3.2 lying on the three sides of P through a given vertex.

5 Two properties of X

The following two statements were found experimentally and verified to high precision; we have no synthetic proof of either.

Theorem 5.1. *X lies on the line* QA-P3 $\cdot$ QA-P4. *Equivalently,*

$$X = \text{QA-P2} \cdot \text{QA-P23} \;\cap\; \text{QA-P3} \cdot \text{QA-P4},$$

and, since QA-P4(P^*) = QA-P4 *(also verified numerically), the Gergonne–Steiner point of P lies on the line* QA-P2(P^*) $\cdot$ QA-P4(P^*)*, the line* QA-L2 *of P^*.*

The ratio in which X divides QA-P3 QA-P4 is not constant, so Theorem 5.1 is not a consequence of an affine relation between the four points involved; we checked in particular that X is neither the reflection of QA-P4 in QA-P3 nor is QA-P23 the reflection of QA-P6 in QA-P1.

Lemma 5.2 (companion point of a concurrent family)**.** *Let $C_1, \dots, C_4$ be four circles through a common point Z, in general position, and let V_{ij} be the second intersection of C_i and C_j. Then the four circles $D_i = \odot(V_{jk}V_{jl}V_{kl})$ have a common point Z^*, and the four inverses of Z^* in the circles C_i lie on one circle through Z.*

Proof. Let ι be an inversion with centre Z. The images $\ell_i = \iota(C_i)$ are four lines, $\ell_j \cap \ell_k = \iota(V_{jk})$, and $\iota(D_i)$ is the circumcircle of the triangle bounded by ℓ_j, ℓ_k, ℓ_l. By Miquel's theorem those four circumcircles have a common point μ, whence the D_i have the common point $Z^* = \iota(\mu)$. For the second statement, $\iota \circ \text{inv}_{C_i} \circ \iota = \text{refl}_{\ell_i}$, so the images of the four inverses are the reflections of μ in the four lines; by Steiner's theorem they are collinear, and the pre-image of that line is a circle through Z. □

Applied to the four pedal circles Π_i of P_i with respect to Δ_i — which pass through QA-P2 [3, QA-P2] — Lemma 5.2 produces a companion point of QA-P2. By Corollary 3.2 the second intersection of Π_i and Π_j is the midpoint of $P_i^*P_j^*$: indeed Π_i is also the pedal circle of P_i^*, so it meets the line P_kP_l at the feet of the perpendiculars from P_i and from P_i^*, and the latter foot is the midpoint of $P_i^*P_j^*$, which lies on Π_j for the same reason. Hence D_i is the nine-point circle of $P_j^*P_k^*P_l^*$ and the companion point is X.

Theorem 5.3. *The four inverses of X in the pedal circles Π_i lie on one circle, which passes through* QA-P2 *and through* QA-P4.

The concyclicity and the incidence with QA-P2 are Lemma 5.2; the incidence with QA-P4 is the experimental part.

6 Verification

All statements were verified numerically on 200 random quadrangles by the accompanying program `verify/verify_isogonal_quadrangle.py`; the largest relative discrepancy is $8.7 \cdot 10^{-14}$ for Theorem 2.1 and its corollaries, $2.6 \cdot 10^{-10}$ for Theorem 5.1 and $7.1 \cdot 10^{-10}$ for Theorem 5.3. The point X is not one of the 44 registered quadrangle centres of [3], and the crosspoint QA-P2 $\cdot$ QA-P23 $\cap$ QA-P3 $\cdot$ QA-P4 does not occur in the list QA-5 of 55 non-registered crosspoints there.

References

[1] R. A. Johnson, *Advanced Euclidean Geometry*, Dover, 1960 (reprint of *Modern Geometry*, 1929).

[2] H. S. M. Coxeter and S. L. Greitzer, *Geometry Revisited*, Mathematical Association of America, 1967.

[3] C. van Tienhoven, *Encyclopedia of Quadri-Figures*, `https://www.chrisvantienhoven.nl/epg/n-geometry/quadri-figures/`; items QA-P1, QA-P2, QA-P3, QA-P4, QA-P5, QA-P12, QA-P23, QA-P32, QA-L1, QA-5 (list of crosspoints). The result that QA-P23 is the centroid of the quadrangle of the isogonal conjugates is due to B. Scimemi (2 October 2015) and is recorded on the page QA-P23.